

WEEK 10

Week 10A — Inodes and File Links

This program demonstrates how to inspect **inode numbers** and file information using `stat()`.

```
GNU nano 8.7.1 week10a.c *
#include <stdio.h>
#include <sys/stat.h>

int main()
{
    struct stat file;

    if (stat("sample.txt", &file) == -1)
    {
        printf("File not found\n");
        return 1;
    }

    printf("File: sample.txt\n");
    printf("Inode Number: %lu\n", file.st_ino);
    printf("File Size: %ld bytes\n", file.st_size);
    printf("File Permissions: %o\n", file.st_mode & 0777);

    return 0;
}
```

```
sri_dhruthi@Dhruthi:~$ nano week10a.c
sri_dhruthi@Dhruthi:~$ nano sample.txt
sri_dhruthi@Dhruthi:~$ gcc week10a.c -o week10a
sri_dhruthi@Dhruthi:~$ ./week10a
File: sample.txt
Inode Number: 24405
File Size: 106 bytes
File Permissions: 644
sri_dhruthi@Dhruthi:~$ ls -li sample.txt
24405 sample.txt
sri_dhruthi@Dhruthi:~$ stat sample.txt
  File: sample.txt
  size: 106          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 24405       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/sri_dhruthi)   Gid: ( 1000/sri_dhruthi)
Access: 2026-09-29 14:43:39.215012636 +0000
Modify: 2026-09-30 14:10:43.962320340 +0000
Change: 2026-09-30 14:10:43.962320340 +0000
 Birth: 2026-09-29 14:42:31.248186474 +0000
sri_dhruthi@Dhruthi:~$
```

Week 10B — Memory-Mapped File Using mmap()

This program uses mmap() to map a file into the process's memory space.

```
sri_dhruthi@Dhruthi: ~
GNU nano 8.7.1 week10b.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>

int main()
{
    int fd;
    struct stat st;
    char *data;

    fd = open("sample.txt", O_RDONLY);

    if (fd < 0)
    {
        printf("File opening failed\n");
        return 1;
    }

    if (fstat(fd, &st) == -1)
    {
        printf("Cannot get file information\n");
        close(fd);
        return 1;
    }

    data = mmap(NULL, st.st_size, PROT_READ,
                MAP_PRIVATE, fd, 0);

    if (data == MAP_FAILED)
    {
        printf("mmap failed\n");
        close(fd);
        return 1;
    }
}
```

```
sri_dhruthi@Dhruthi:~$ nano week10b.c
sri_dhruthi@Dhruthi:~$ gcc week10b.c -o week10b
sri_dhruthi@Dhruthi:~$ ./week10b
File contents using mmap():
Operating System Practical
File Copy using System Calls
open read write close
This file is used as input.
```

